

Based on what we've covered in our previous sessions, here are your **complete HTML & CSS revision notes**.

HTML & CSS Notes

HTML (HyperText Markup Language)

HTML is used to **create the structure of a webpage**.

Example:

```
<!DOCTYPE html>
<html>
<head>
  <title>My Website</title>
</head>
<body>

  <h1>Hello World</h1>
  <p>Welcome to HTML.</p>

</body>
</html>
```

HTML Document Structure

```
<!DOCTYPE html>
<html>
<head>
  <title>Page Title</title>
</head>
<body>

</body>
</html>
```

Tag	Purpose
<code><!DOCTYPE html></code>	Tells browser HTML5 is used
<code><html></code>	Root element
<code><head></code>	Metadata
<code><title></code>	Browser tab title
<code><body></code>	Visible content

Headings

`<h1>Main Heading</h1>`
`<h2>Sub Heading</h2>`
`<h3>Heading</h3>`
`<h4>Heading</h4>`
`<h5>Heading</h5>`
`<h6>Heading</h6>`

Largest → h1

Smallest → h6

Paragraph

`<p>This is a paragraph.</p>`

Lists

Ordered List

```
<ol>
  <li>Apple</li>
  <li>Orange</li>
</ol>
```

Output

1. Apple
 2. Orange
-

Unordered List

```
<ul>
  <li>Apple</li>
  <li>Orange</li>
</ul>
```

Output

- Apple
 - Orange
-

Links

```
<a href="https://google.com">
Google
</a>
```

Open in new tab

```
<a href="https://google.com" target="_blank">
Google
</a>
```

Images

```

```

Resize

```

```

Div

A `<div>` is a container.

```
<div>
```

```
<h1>Title</h1>
```

```
<p>Paragraph</p>
```

```
</div>
```

Used for layout.

Tables

```
<table>
```

```
<tr>
```

```
<th>Name</th>
```

```
<th>Age</th>
```

```
</tr>
```

```
<tr>
```

```
<td>John</td>
```

```
<td>20</td>
```

```
</tr>
```

</table>

Table Tags

Tag	Meaning
-----	---------

table	Creates table
-------	---------------

tr	Table row
----	-----------

th	Header
----	--------

td	Data
----	------

Common Table Attributes (HTML)

Attribute	Purpose
colspan	Merge columns
rowspan	Merge rows
border (<i>older HTML</i>)	Border
cellpadding (<i>older HTML</i>)	Space inside cells
cellspacing (<i>older HTML</i>)	Space between cells

In modern HTML, borders and spacing are usually controlled with CSS instead of HTML attributes.

Example

```
<td colspan="2">
Merged
</td>
```

Forms (Basic)

```
<form>
```

```
<input type="text">
```

```
<input type="password">
```

```
<input type="email">
```

```
<button>Submit</button>
```

```
</form>
```

CSS (Cascading Style Sheets)

CSS styles webpages.

Example

```
h1{  
  color:red;  
}
```

Ways to Add CSS

Inline

```
<h1 style="color:red;">  
Hello  
</h1>
```

Internal

```
<style>
```

```
h1{  
color:red;  
}
```

```
</style>
```

External (Best)

HTML

```
<link rel="stylesheet" href="style.css">
```

CSS

```
h1{  
color:red;  
}
```

CSS Syntax

```
selector{  
  
property:value;  
  
}
```

Example

```
p{  
  
color:blue;  
  
}
```

Selectors

Element

```
p{  
  color:red;  
}
```

Class

HTML

```
<p class="text">
```

CSS

```
.text{  
  
  color:blue;  
  
}
```

ID

HTML

```
<p id="title">
```

CSS

```
#title{  
  
  color:red;  
  
}
```

Colors

`color:red;`

`color:#ff0000;`

`color:rgb(255,0,0);`

Background

`background-color:black;`

Width

`width:500px;`

Height

`height:300px;`

Margin

Outside spacing

`margin:20px;`

Center element

`margin:auto;`

Padding

Inside spacing

```
padding:20px;
```

Border

```
border:2px solid black;
```

Text Align

```
text-align:center;
```

Display

```
display:block;
```

```
display:inline;
```

```
display:flex;
```

```
display:grid;
```

Flexbox

Flexbox arranges items in one direction (row or column).

```
.container{  
  
display:flex;  
  
}
```

justify-content

Horizontal alignment (default row)

```
justify-content:center;
```

Options

- flex-start
 - center
 - flex-end
 - space-between
 - space-around
 - space-evenly
-

align-items

Vertical alignment (default row)

```
align-items:center;
```

Options

- flex-start
 - center
 - flex-end
 - stretch
-

flex-direction

```
flex-direction:row;
```

```
flex-direction:column;
```

flex-wrap

```
flex-wrap:wrap;
```

order

Changes visual order.

```
.item{
```

```
order:2;
```

```
}
```

align-self

Align one item differently.

```
align-self:flex-end;
```

Centering a Div

```
body{
```

```
margin:0;
```

```
height:100vh;
```

```
display:flex;
```

```
justify-content:center;
```

```
align-items:center;

}
```

Grid

Grid is used for **2D layouts (rows and columns)**.

```
.container{

display:grid;

grid-template-columns:1fr 1fr;

gap:20px;

}
```

Grid Properties

grid-template-columns

```
grid-template-columns:
1fr 1fr 1fr;
```

grid-template-rows

```
grid-template-rows:
100px 200px;
```

gap

```
gap:20px;
```

column-gap

column-gap:20px;

row-gap

row-gap:20px;

Flex vs Grid

Flexbox	Grid
1D Layout	2D Layout
Row OR Column	Rows AND Columns
Navigation bars	Full page layouts
Simpler layouts	Complex layouts
Content-driven	Layout-driven

CSS Units

px
%
vh
vw
em
rem
fr

Common CSS Properties

color
background-color

width
height
margin
padding
border
display
text-align
font-size
font-family
justify-content
align-items
flex-direction
flex-wrap
order
align-self
grid-template-columns
grid-template-rows
gap

Example (Table Centered)

HTML

```
<table>
<tr>
<td>Hello</td>
</tr>
</table>
```

CSS

```
table{
  width:500px;
  margin:auto;
}
```

```
body{
  margin:0;
  height:100vh;
```

```
display:flex;  
justify-content:center;  
align-items:center;  
}
```
